Rotary Wing Aerodynamics (AE 667)



Tail Rotor Design

Functions

- Provide anti torque reaction
- Provide yaw control.
- Provides yaw stability.



Design Parameters

- Diameter / Disk Loading
- Tip speed
- Number of blades
- Twist
- Pusher or Tractor?
- Direction of rotation
- Location

Design Parameters

- Diameter/Disk Loading
  - Large diameter/Low Disk Loading: Reduces power required
  - Small diameter/High Disk Loading: Lower weight; smaller problems with helicopter c. g. position; better ground clearance; better clearance with main rotor, horizontal stabilizer and tail boom.
  - Aim for smallest possible diameter that meets all design requirements.
- Tip speed
  - Low tip speed : minimizes noise and compressibility effects. Requires higher torque.
  - High tip speed: minimizes component weight and stall effects. Higher noise, higher control forces.

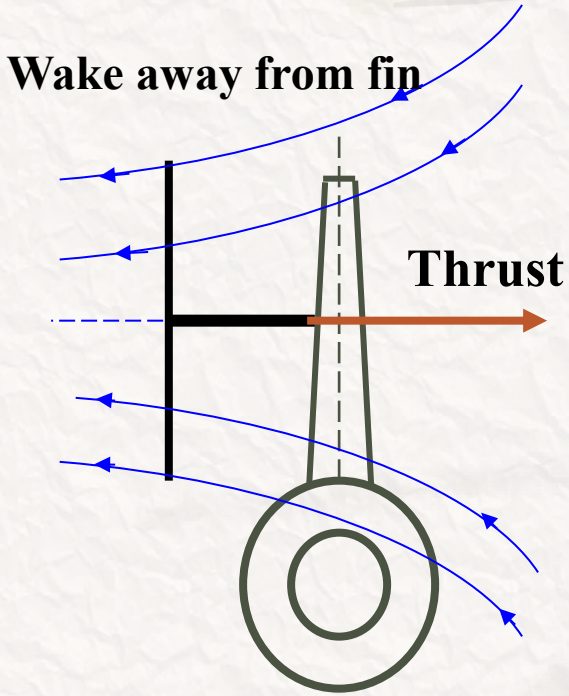
Design Parameters

- Number of blades
  - Fewer blades: cheaper to build and maintain
    stubby blades & higher tip losses
  - Thumb rules:
    - Blade aspect ratio between 5 and 9.
    - Solidity between 0.03 and 0.06.

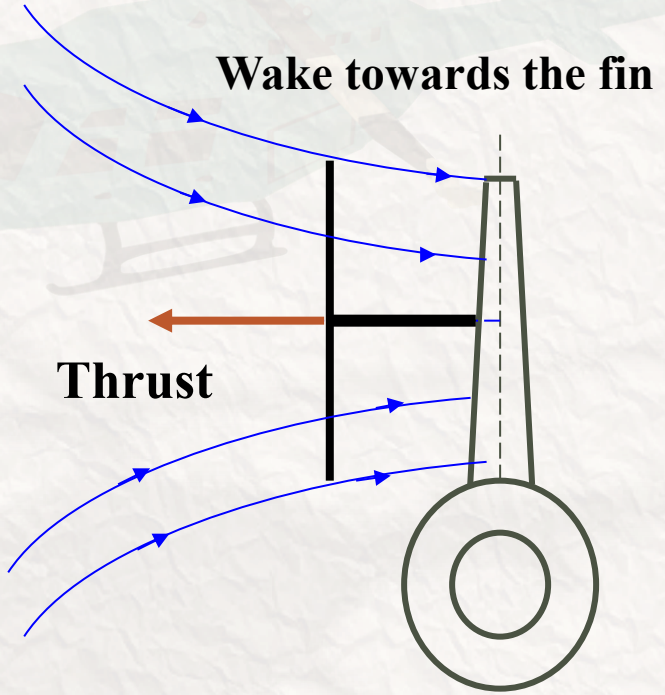
- Twist
  - Similar to main rotor blade.
  - High twist is good for hover, but penalizes forward flight performance.
  - < 10 degrees twist is used.

Design Parameters

- Pusher or Tractor?
  - Net thrust on tractor less than on pusher.



PUSHER



TRACTOR

Design Parameters

- Direction of rotation :“AFT AT THE TOP”
  - Why?
  - Found to be more effective…
  - Main-rotor tail-rotor interaction effect



Design Parameters

- Tail rotor position w.r.t main rotor and C.G:

Special Cases			
Type of Main Rotor	Teetering	Any	Very High Offset
Height of Tail Rotor	As High as Main	As Low as CG	Any
Results	Fuselage Level	Fuselage Tilted, No Flapping	Fuselage Tilted, Very Little Flapping

